

SCS (Order Management System) Use Case Document

UC1: Process Customer Sale at Checkout

Actors: Retail Associate (**Primary**), Customer (**Secondary**), System

Preconditions:

- Product Master is up to date with all items, prices, GST rates, reorder thresholds
- System is online and connected to local inventory DB
- POS terminal is operational

Main Flow:

1. Customer arrives at checkout counter with items
2. Associate scans the first item's barcode
3. System decodes GTIN/EAN and queries Product Master
4. System retrieves product name, price, GST rate, current stock
 - a. If item is found: proceed to step 5
 - b. if item barcode is unreadable: refer to Alternative flow A
 - c. If item not found: refer to E1 Flow
5. Process Item Based on Type:
 - a. For packaged/scannable items: System adds item with pre-stored quantity
 - b. For loose/variable-weight items: Associate places item on scale, system reads weight, calculates extended price (unit price × weight)
 - c. System displays item, quantity/weight, and price on POS
6. Associate repeats steps 2-5 for remaining items
7. System calculates subtotal, total GST, and final bill amount
8. System displays total and asks for payment method (Cash / Digital / Credit)
9. [Branch: Cash Payment Flow] or [Branch: Digital Payment Flow]

a. Cash Payment Flow:

- i. Associate collects cash from customer
- ii. If cash amount > bill total: System calculates change and displays it
- iii. Associate hands over change to customer
- iv. Associate marks payment as "Paid - Cash" in system
- v. Proceed to next step

b. Digital Payment:

- i. Associate launches payment gateway (Card/UPI)
- ii. Customer completes payment
- iii. System receives confirmation
- iv. System marks payment as "Paid - Digital [Sub-type: Card/UPI]"
- v. Proceed to next step

c. Credit: [Refer Alternative flow B]

10. After payment confirmed: System deducts inventory for all items

11. For each item where stock now $\leq$ reorder threshold: System generates low-stock alert

12. System generates receipt with order number, timestamp, items, total cost

13. Associate hands receipt to customer

14. System records order in Orders table with complete details

Postconditions:

- Order is recorded in system with all details
- Inventory is updated in real-time
- Low-stock alerts are sent (if applicable)
- Receipt is printed and given to customer

Alternative Flows:

A. Barcode unreadable:

- i. Associate manually enters product code or name
- ii. System searches Product Master
- iii. Proceed to step 5

B. Credit (Udhaar) Payment [Credit Ledger Management: OUT OF SCOPE]:

- i. Associate selects "Credit" as the payment type in the system
- ii. System marks the order as "Paid - Credit" and generates a receipt with order ID
- iii. Associate notes the order ID in the manual Credit Register for external tracking
- iv. Proceed to inventory deduction step

Note: The system treats the order as fully completed. All downstream credit tracking and collection is managed manually outside the OMS.

C. Payment Declined:

- i. System notifies associate of declined payment
- ii. Associate attempts alternative payment method
- iii. If all methods fail: Cancel order, no inventory deduction

Exceptions:

E1. Item Unavailable or Insufficient Stock:

- Triggered when one of the following occurs:
 - Customer requests a product they cannot locate in the store, and Associate searches by product name in the system and finds stock quantity = 0
 - Customer requests a specific quantity of a product (e.g., 3 units) but system stock shows fewer units available than requested
- Associate communicates the unavailability or insufficient quantity to the customer.

- Customer either:
 - E1a. Selects an alternative product or reduced quantity → Return to Step 2
 - E1b. Cancels the item or entire order → End use case (no inventory deduction, no order recorded)

UC2: Process Replenishment with the Distributor

Actors: Store Owner (**Primary**), Distributor (**Secondary**), System

Preconditions:

- Product Master is up to date with min_count and max_count thresholds for each item
- Distributor master data (contact info, preferred order method) is configured
- Store Owner is logged into the system
- Inventory has been decremented from a sale (triggering the low-stock condition)

Main Flow:

1. System Detects Low Inventory & Creates Draft PO for the particular Distributor associated for that product and the PO Draft is categorised based on the Distributor:
 - a. System checks if any product quantity $\leq$ min_count threshold
 - b. For each product below threshold:
 - i. Generates unique PO Order ID for that distributor
 - ii. Creates draft PO record with: Product ID, reorder quantity (up to max_count), current date/time
 - iii. Stores in Draft PO table with status "Pending Approval"
2. System generates a low-stock alert and notifies Store Owner (via system dashboard, email, or SMS). Store Owner accesses the Dashboard and reviews the Draft PO table showing:
 - a. All products below minimum threshold
 - b. Current quantity, reorder quantity, Distributor, and date/time of threshold breach for particular product in the draft PO product list

- c. Related metadata (reorder pack size, Distributor contact, etc.)
- 3. Draft POs are organized by Distributors on the Dashboard. For each Distributor's PO, Store Owner can review items, quantities, and estimated cost.
- 4. Store Owner clicks "Approve & Send" for each Distributor's PO. System sends the PO details to the Distributor via [WhatsApp API / Email / SMS] with:
 - a. PO Order ID
 - b. Item list with quantities
 - c. Delivery location
 - d. Total cost
- 5. Delivery Received & Verification
 - a. Store Owner (or delegated Associate) receives goods from Distributor
 - b. Physically verifies item count and condition against the PO
 - c. Scans each product barcode (or manually enters code for non-scannable items) to confirm receipt
 - d. Places verified items on shelves
 - e. Proceeds to Step 6 (GRN Submission) once verification is complete
- 6. GRN Submission & Inventory Update:
 - a. Once delivery verification is complete (Step 5), Store Owner locates the corresponding PO in the system and clicks "Submit GRN" (Goods Receipt Note) for the PO
 - b. System updates Product Master with received quantities (inventory now available for sale)
 - c. System marks PO status as "Completed"
 - d. Low-stock alert is cleared for those products

Postconditions:

- PO is recorded with status "Completed" or "Rejected"
- Product Master inventory quantities are updated (if GRN submitted) or

unchanged (if rejected)

- Low-stock alert is cleared for that product
- Note: Distributor payment tracking remains in manual Distributor Register [outside OMS scope]

Alternative Flow:

A. Stockout During Pending PO:

- i. While a PO for a low-stock item is pending delivery, customer requests that item
- ii. [Reference UC1-E: Item Out of Stock flow]
- iii. Associate communicates unavailability
- iv. After PO is received and GRN submitted, item becomes available again

B. Partial Fulfilment:

- i. If Distributor delivers only part of the ordered quantity:
 1. Store Owner verifies partial delivery quantity
 2. System allows GRN submission for received items only
 3. System updates inventory with partial quantity
 4. System auto-generates a follow-up PO for remaining items (or prompts Store Owner)

C. PO Cancellation by Store Owner

- i. If Store Owner needs to cancel a pending PO (e.g., changed mind on quantities):
 1. Store Owner clicks "Cancel PO" before Distributor confirms delivery
 2. System marks PO as "Cancelled"
 3. Distributor is notified (via system notification or manual message)
 4. No inventory deduction occurs